Connect Bluetooth module to amplifier. Bluetooth module has four output pins, namely, Lout, Rout, Vcc and GND. Right channel (Rout) and left channel (Lout) are connected to pins 2 and 7 of IC2, respectively, followed by presets VR1 and VR2, and capacitors C3 and C5. Pins 3 and 6 of IC2 are connected to feedback capacitors C4 and C6, which are grounded. Bluetooth module is given a power supply of 5V DC.

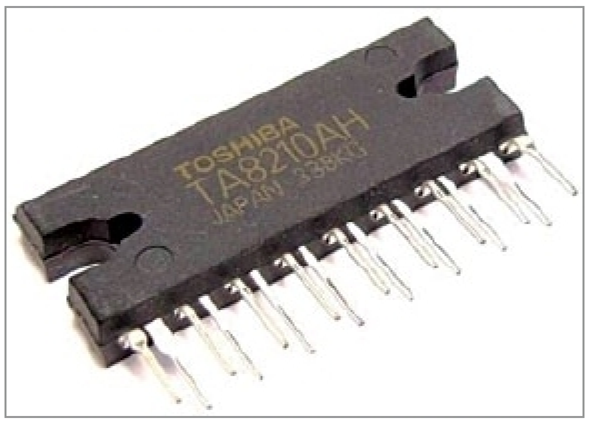

Fig. 4: 2-channel audio amplifier IC

Connect amplifier to speakers. Output pins 11 and 12 of amplifier IC2 are directly connected to a loudspeaker (LS2). Similar configuration is done with output pins 15 and 16 of IC2 and another loudspeaker (LS1). A capacitor in series with a resistor is connected to each wire for preventing audio signal oscillation.

Connect power supply to the circuit. Pins 5, 13 and 14 of IC2 are grounded, while pin 8 is connected to ground through C7, which acts as a ripple-reduction element. Pins 1 and 4 of IC2 are connected to a power supply of 5V. Capacitor C1 is connected to 12V supply line for filtering. Pins 9, 10 and 17 of IC2 are connected to a regulated 12V DC power supply.

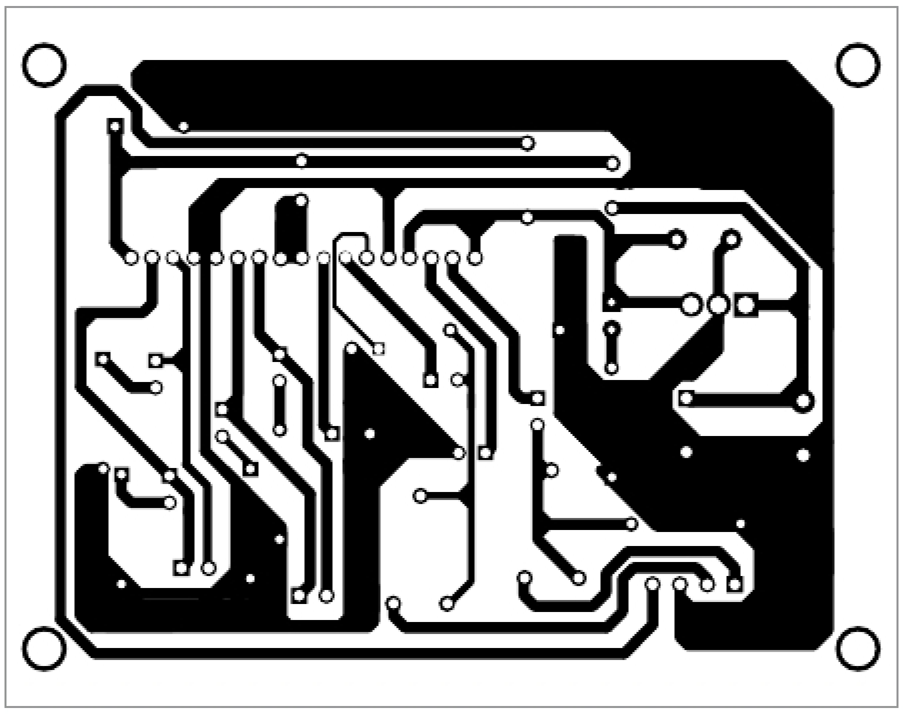

Fig. 5: Actual-size PCB layout of 2-channel audio amplifier

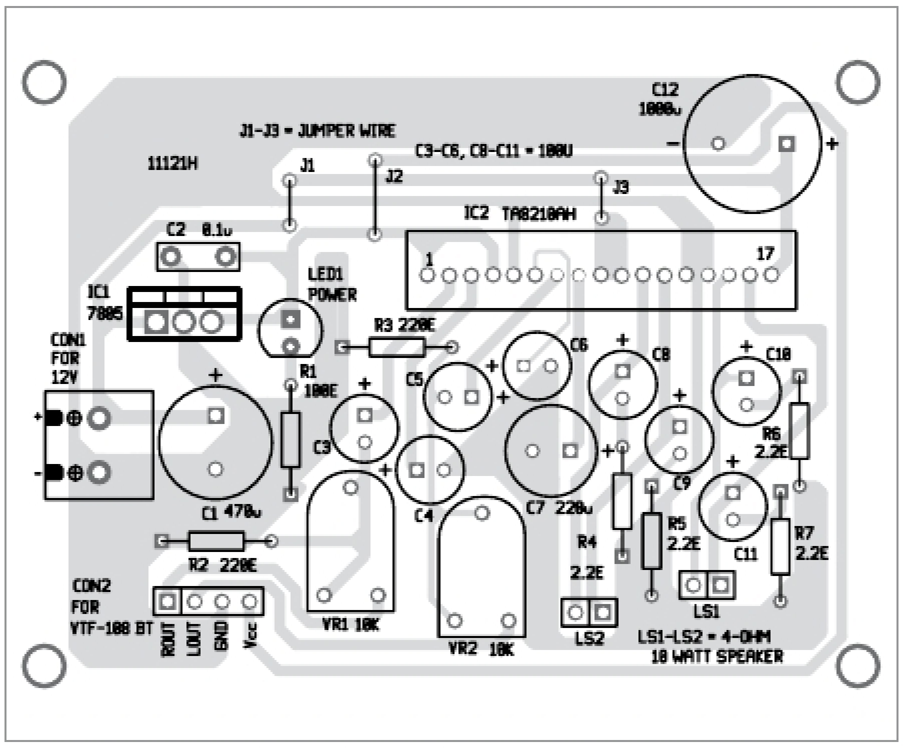

Fig. 6: Components layout for the PCB

Construction and testing

An actual-size PCB layout for the 2-channel audio amplifier is shown in Fig. 5 and its components layout in Fig. 6.

Connecting mobile phone with Bluetooth. The first step is to connect Bluetooth of the mobile phone to Bluetooth module (VTF-108) of the audio amplifier. For this, power on the circuit and turn on Bluetooth from Settings of your mobilephone and pair it with VTF-108.

Pagaria is the name of the Bluetooth device (used for testing purposes) that will appear in your mobile phone. Select it, and it will connect automatically. No password is required for pairing. After pairing, you can start sending audio signals to VTF-108.

Sending audio signals to amplifier. Once audio signals are received by VTF-108, these will be transferred to IC2 through its output pins, Lout and Rout. Adjust VR1 and VR2 until you get maximum and noise-free sound from the speakers.

Amplified signal in the speakers. Amplified audio signal from IC2 is transmitted to speakers LS1 and LS2.

VTF-108 module has a USB port and usually comes with a remote. You can use this remote to change the song or adjust the volume. You can also interface a USB device with this Bluetooth module to play songs. EFY

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2	- TA8210AH, 2-channel audio amplifier
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 100-ohm
R2-R3	- 220-ohm
R4-R7	- 2.2-ohm
VR1-VR2	- 10-kilo-ohm preset
Capacitors:	
C1	- 470µF, 35V electrolytic
C2	- 0.1µF ceramic disk
C3-C6, C8-C11	- 100µF, 25V electrolytic
C7	- 220µF, 16V electrolytic
C12	- 1000µF, 25V electrolytic
Miscellaneous:	
CON1	- 2-pin terminal connector for 12V power supply
CON2	- 4-pin connector
LS1-LS2	- 4-ohm, 10-watt speaker
	- Heat-sink for IC1 and IC2
	- 12V DC, 2A power adaptor or 12V battery
	- Mobile phone with inbuilt Bluetooth and songs in memory
	- Jumper wires
	- VTF-108 Bluetooth module

Vasavi Sai VPDK (left) is editorial member of International Journal of Information Technology. His interests include LED fabrication and designing, embedded systems, home automation, VLSI and FPGA. He has published research papers in various international journals. Saurabh Kumar (right) is currently chief administrative officer at Doorstepshoppee. His areas of interest are embedded systems and MATLAB programming